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Topic 4 — Extracting Metals and Equilibria

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 1

This topic has two halves. The first half is about **metals**: how reactive they are, why some are found as the pure element while most are locked up in ores, and how the position of a metal in the **reactivity series** decides how we get it out of the ground. It finishes with the bigger picture — biological extraction, recycling and **life cycle assessments**. The second half introduces **reversible reactions** and **dynamic equilibrium**, using the Haber process for ammonia as the main example, and (at Higher tier) shows how changing the conditions shifts the position of an equilibrium.

Content marked **(HT)** is Higher Tier only. All of Topic 4 is examined in both GCSE Chemistry (1CH0) and Combined Science; the chemistry-only extension of the Haber process, fertilisers and yield/atom economy is in Topic 5 (Separate Chemistry 1).

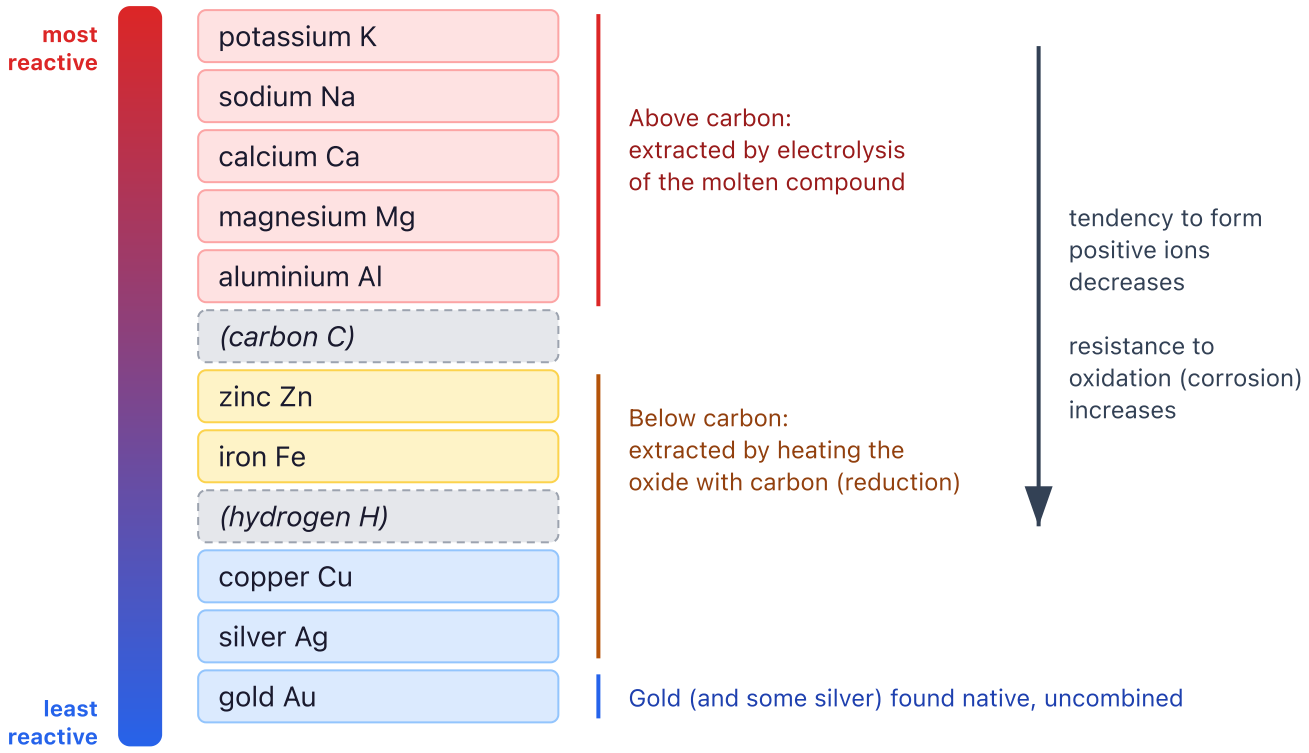
4.1 The Reactivity Series

> The **reactivity series** is a list of metals arranged in order of how vigorously they react. A metal high in the series reacts readily because its atoms have a strong tendency to **lose electrons and form positive ions**. Carbon and hydrogen are non-metals but are included as reference points.

The order you need for Edexcel is:

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The reactivity series (Edexcel 1CH0)



> **Exam Tip:** A popular mnemonic: **P**lease **S**top **C**alling **M**e **A** **C**areless **Z**ebra, **I**nstead **H**elp **C**ount **S**ome **G**iraffes — K, Na, Ca, Mg, Al, C, Zn, Fe, H, Cu, Ag, Au. Lithium is not in the Edexcel list but, if it appears, it sits between sodium and calcium.

Deducing an order of reactivity from experiments

You need to be able to put metals in order using three kinds of evidence: their reactions with **water**, with **dilute acid**, and **displacement reactions** with solutions of other metal salts. The more vigorous the reaction (more bubbles, faster temperature rise, metal disappears more quickly), the more reactive the metal.

Reactions with water

Reactive metals react with cold water to form a **metal hydroxide** and **hydrogen**:

metal + water → metal hydroxide + hydrogen

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$$

Less reactive metals such as magnesium, zinc and iron react only very slowly with cold water but react with **steam** to form a **metal oxide** and hydrogen:

$$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$$

Copper, silver and gold do not react with water or steam.

Reactions with dilute acids

Metals above hydrogen react with dilute hydrochloric or sulfuric acid to produce a **salt** and **hydrogen**. Metals below hydrogen (copper, silver, gold) do not react with dilute acids.

$$\text{metal} + \text{acid} \rightarrow \text{salt} + \text{hydrogen}$$

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$

The hydrogen can be confirmed with a **lighted splint**, which burns with a squeaky pop.

Metal	Cold water	Dilute hydrochloric acid
Potassium	Floats, melts into a ball, fizzes violently, hydrogen ignites with a lilac flame	Explosive — never done in school
Sodium	Floats, melts, fizzes rapidly, moves around the surface	Explosive — never done in school
Calcium	Sinks, fizzes steadily, solution goes cloudy ($Ca(OH)_2$ is only slightly soluble)	Very vigorous

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Magnesium	A few bubbles after a long time; reacts well with steam	Rapid fizzing, gets hot, ribbon disappears quickly
Zinc	No visible reaction (reacts with steam)	Steady bubbling
Iron	Rusts slowly over days (reacts with steam when red-hot)	Slow bubbling
Copper	No reaction	No reaction

Exam Tip: Aluminium often seems less reactive than its position suggests because it is covered by a thin, tough layer of **aluminium oxide** that stops water and acid reaching the metal. If you are asked why, say the oxide layer is *protective* — not that aluminium is unreactive.

Making it a fair test

When comparing metals with acid, keep the following the same: the **concentration** and **volume** of acid, the **temperature**, and the **mass and surface area** of metal (e.g. use powders of similar particle size, or same-length strips). Measure something objective, such as the **temperature rise** in a fixed time or the **volume of hydrogen** collected in a gas syringe in the first minute.

Worked Example: A student adds 0.5 g of four metal powders, W, X, Y and Z, to 25 cm^3 of 1 mol dm^{-3} hydrochloric acid in turn and records the temperature rise.

W: +2 °C X: +0 °C Y: +18 °C Z: +7 °C

The greater the temperature rise, the more energy released and the more vigorous the reaction. Order (most → least reactive): **Y > Z > W > X**. X may be copper (below hydrogen, no reaction). This is consistent with Y = magnesium, Z = zinc, W = iron.

4.2 Displacement Reactions

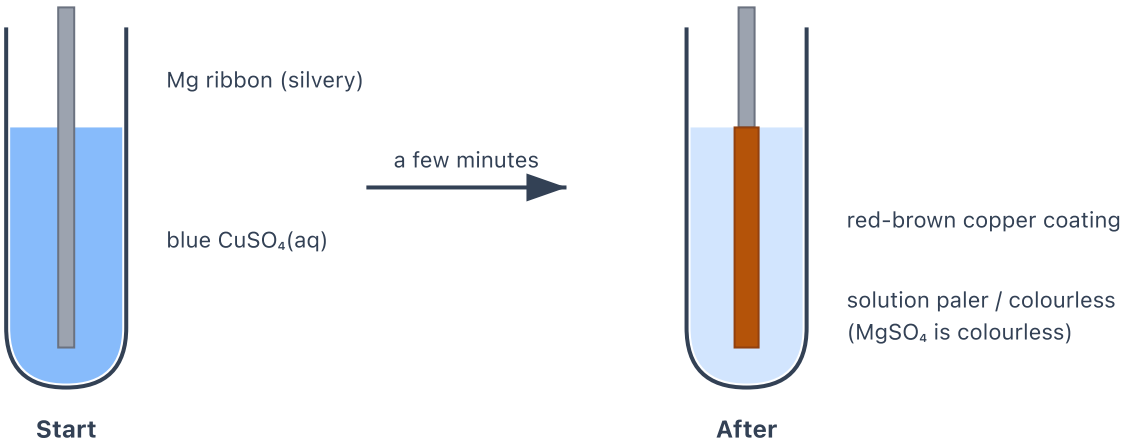
In a **displacement reaction** a more reactive metal takes the place of a less reactive metal in one of its compounds. A metal will displace any metal **below** it in the reactivity series from a solution of its salt (or from its oxide when heated).

magnesium + copper sulfate → magnesium sulfate + copper

$$Mg(s) + CuSO_4(aq) \rightarrow MgSO_4(aq) + Cu(s)$$

Observations: the blue colour of the copper(II) sulfate solution fades, a red-brown/pink coating of copper forms on the magnesium, and the mixture warms up (the reaction is exothermic). If the metal added is *less* reactive than the one in solution (e.g. copper in magnesium sulfate) there is no reaction.

Magnesium ribbon in copper(II) sulfate solution



Using displacement to rank metals

A grid of results is a quick way to deduce a reactivity order: the metal that displaces the most others is the most reactive.

Metal added ↓ / solution →	magnesium sulfate	zinc sulfate	iron(II) sulfate	copper(II) sulfate	Reactions
Magnesium	—	✓	✓	✓	3
Zinc	×	—	✓	✓	2

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Iron	×	×	—	✓	1
Copper	×	×	×	—	0

The thermite reaction

Displacement also works with solid oxides. Aluminium displaces iron from iron(III) oxide in a very exothermic reaction that produces molten iron — used to weld railway lines:

$$2Al(s) + Fe_2O_3(s) \rightarrow Al_2O_3(s) + 2Fe(l)$$

Displacement as a redox reaction (HT)

In terms of electrons, the more reactive metal **loses electrons** (is **oxidised**) and the ion of the less reactive metal **gains electrons** (is **reduced**). Because oxidation and reduction happen together, displacement reactions are **redox** reactions.

> **Oxidation is Loss** of electrons; **Reduction Is Gain** of electrons — **OIL RIG**.

The sulfate ions do not change; they are **spectator ions**. Leaving them out gives the **ionic equation**, which can be split into two **half equations**:

$$\text{Ionic: } Mg(s) + Cu^{2+}(aq) \rightarrow Mg^{2+}(aq) + Cu(s)$$

$$\text{Oxidation: } Mg \rightarrow Mg^{2+} + 2e^- \quad | \quad \text{Reduction: } Cu^{2+} + 2e^- \rightarrow Cu$$

Worked Example: (HT) Zinc powder is added to silver nitrate solution. Write the ionic equation and identify what is oxidised and reduced.

Full equation: $Zn(s) + 2AgNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + 2Ag(s)$
Nitrate ions are spectators. Ionic equation: **$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$**
$Zn \rightarrow Zn^{2+} + 2e^-$ (zinc atoms lose electrons → **zinc is oxidised**)
$2Ag^+ + 2e^- \rightarrow 2Ag$ (silver ions gain electrons → **silver ions are reduced**)

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Each zinc atom loses 2 electrons but each silver ion gains only 1, so **two** Ag^+ ions are needed per Zn atom — the electrons must balance.

Exam Tip: When you say what is oxidised or reduced, name the exact particle: "zinc **atoms** are oxidised" and "silver **ions** are reduced". "Silver nitrate is reduced" usually loses the mark.

4.3 Why Metals Differ in Reactivity

Metals react by **losing their outer electrons** to form positive ions. The reactivity series is really a list of how easily each metal does this:

- Potassium forms K^+ very easily, so it reacts violently with water.
- Copper has only a weak tendency to form Cu^{2+}, so it does not react with water or dilute acids.
- Gold has almost no tendency to form ions, which is why it stays shiny for thousands of years.

Resistance to oxidation

Because the less reactive metals have a low tendency to form positive ions, they are **more resistant to oxidation**. This explains:

- **Gold and silver** are used for jewellery and electrical contacts: they do not tarnish (or tarnish only slowly).
- **Copper** is used for water pipes: it does not react with water.
- **Iron** rusts (is oxidised) quite easily and must be protected.
- **Potassium and sodium** oxidise in air within seconds and must be stored under oil.

4.4 Oxidation and Reduction — the Oxygen Definition

Oxidation is the **gain of oxygen** by a substance.
Reduction is the **loss of oxygen** from a substance.

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A substance that removes oxygen from another is a **reducing agent** (e.g. carbon, carbon monoxide).

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s) \quad \text{(magnesium is oxidised)}$$

$$2CuO(s) + C(s) \rightarrow 2Cu(s) + CO_2(g) \quad \text{(copper oxide is reduced; carbon is oxidised)}$$

In the second reaction, copper(II) oxide loses oxygen (reduced) while carbon gains oxygen (oxidised). Oxidation and reduction always occur together.

Exam Tip: At Foundation you only need the oxygen definition. At Higher you may be asked to use *either* definition — read the question carefully. For reactions involving ions (displacement in solution, electrolysis) the **electron** definition is the one that earns marks.

4.5 Ores and the Extraction of Metals

An **ore** is a rock containing enough of a metal or metal compound to make it economically worthwhile to extract the metal. Most metals are found in the Earth's crust combined with other elements, typically as **oxides**, **sulfides** or **carbonates**.

Very unreactive metals such as **gold** (and some silver, copper and platinum) are found in the Earth's crust as the **uncombined element** ("native" metal), because they are too unreactive to have formed compounds. These can be separated from rock physically, e.g. by panning.

Metal	Common ore	Main compound
Iron	Haematite	Iron(III) oxide, Fe_2O_3
Aluminium	Bauxite	Aluminium oxide, Al_2O_3
Copper	Chalcopyrite / malachite	$CuFeS_2$ / copper carbonate compounds

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Zinc	Zinc blende	Zinc sulfide, ZnS
Gold	Native gold	Uncombined Au

The method depends on reactivity

Extracting a metal from its oxide means **reducing** it (removing oxygen). The method chosen depends on where the metal is in the reactivity series relative to **carbon**:

Position	Method	Examples	Why
Above carbon	Electrolysis of the molten ore (or molten compound)	K, Na, Ca, Mg, Al	Carbon is less reactive, so it cannot remove the oxygen. Electrolysis is needed but uses large amounts of electricity — expensive.
Below carbon	Heating with carbon (carbon reduction)	Zn, Fe, Cu	Carbon is more reactive, so it displaces the metal from its oxide. Carbon (coke) is cheap and also provides heat when it burns.
Very unreactive	Found native; physical separation	Au, (Ag, Pt)	Already uncombined.

Extracting iron — reduction with carbon

Iron is extracted from haematite in a **blast furnace**. Coke (carbon) burns in hot air to form carbon dioxide; this reacts with more coke to form **carbon monoxide**, which acts as the main reducing agent. The overall idea is simply that carbon removes oxygen from iron(III) oxide:

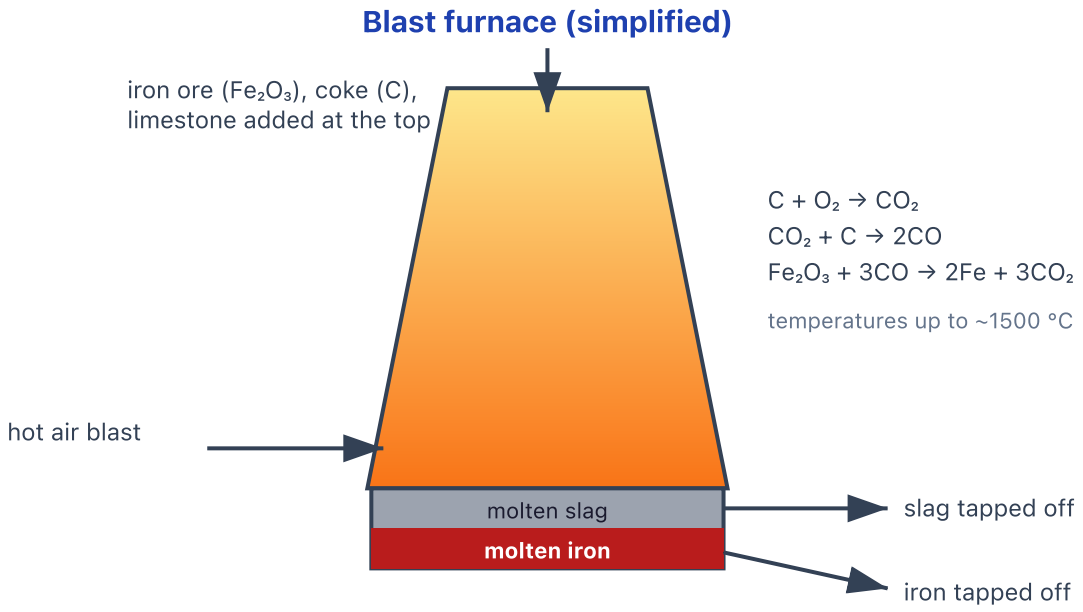
$$2Fe_2O_3(s) + 3C(s) \rightarrow 4Fe(l) + 3CO_2(g)$$

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$$

Iron(III) oxide is **reduced** (loses oxygen); carbon/carbon monoxide is **oxidised** (gains oxygen). Limestone is added to remove acidic impurities (mainly silica) as **slag**.

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Exam Tip: The specification states that knowledge of the blast furnace is **not required** — you only need to know that iron is extracted by heating its oxide with carbon, because iron is below carbon in the reactivity series. The detail here is background.



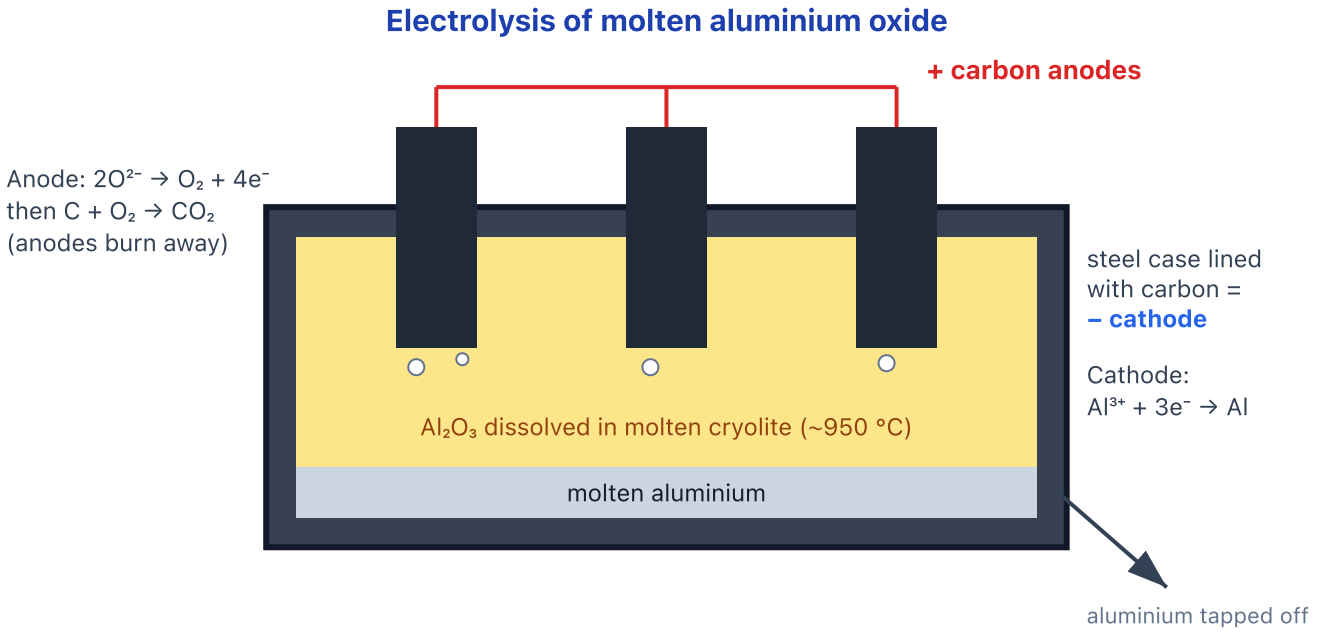
Extracting aluminium — electrolysis

Aluminium is above carbon, so it is extracted by **electrolysis** of aluminium oxide (purified from bauxite). Key points:

- Aluminium oxide has a very high melting point (over 2000 °C). It is dissolved in molten **cryolite**, which lowers the operating temperature to about 950 °C — saving energy and cost.
- The ions must be free to move, so the electrolyte must be molten (or dissolved).
- At the **cathode** (negative electrode) aluminium ions gain electrons (**(HT)** $Al^{3+} + 3e^- \rightarrow Al$). Molten aluminium collects at the bottom.
- At the **anode** (positive electrode) oxide ions lose electrons (**(HT)** $2O^{2-} \rightarrow O_2 + 4e^-$).
- The anodes are made of **carbon (graphite)**. The hot oxygen reacts with them to form carbon dioxide ($C + O_2 \rightarrow CO_2$), so the anodes burn away and must be **replaced regularly**.

Overall: $2Al_2O_3(l) \rightarrow 4Al(l) + 3O_2(g)$

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Exam Tip: Why is aluminium expensive to extract? Give **two** reasons: (1) electrolysis needs a very large amount of **electricity**, and energy is also needed to keep the electrolyte molten; (2) the carbon **anodes must be replaced** because they react with oxygen. Cryolite is used to *reduce* the energy cost, not to make the ions move.

Worked Example: Explain why zinc can be extracted by heating zinc oxide with carbon, but magnesium cannot be extracted from magnesium oxide this way.

Carbon can only remove oxygen from metals that are **less reactive than carbon**. Zinc is below carbon, so carbon reduces zinc oxide: ZnO + C → Zn + CO. Magnesium is **above** carbon — it holds on to oxygen more strongly than carbon does — so carbon cannot reduce magnesium oxide, and magnesium must be extracted by **electrolysis** of a molten compound. In general: **metal below carbon → carbon reduction; metal above carbon → electrolysis**.

4.6 Biological Methods of Extraction (HT)

The world's supply of high-grade ores (especially copper) is running out. **Low-grade ores** contain only a small percentage of metal, so traditional mining and smelting would be uneconomic. Two

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biological methods can be used:

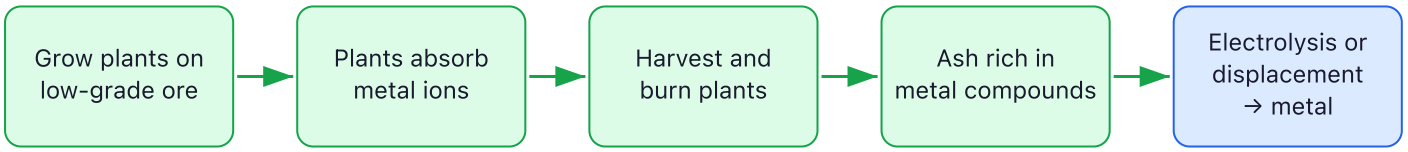
Bacterial extraction (bioleaching)

- Certain bacteria feed on low-grade ores (e.g. copper sulfide) and produce a **leachate** — a solution containing metal ions such as Cu^{2+}.
- The metal is then recovered from the leachate, for example by **displacement with scrap iron** ($Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$) or by electrolysis.

Phytoextraction (phytomining)

- **Plants** are grown on soil containing low concentrations of the metal; they absorb metal ions through their roots and concentrate them in their tissues.
- The plants are harvested and **burned**; the ash contains a high concentration of metal compounds.
- The metal is extracted from the ash (e.g. dissolved in acid then electrolysed or displaced).

Phytoextraction — the steps



Bioleaching is similar but bacteria produce a metal-ion solution (leachate) instead of plants and ash.

Evaluating biological methods

Advantages	Disadvantages
Can extract metal economically from **low-grade ores** and mine waste that would otherwise be useless	Very **slow** — plants take months/years to grow; bacteria take a long time
Less damage to the landscape — no large open-cast mines, less waste rock	Phytoextraction needs large areas of land

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Uses less energy than smelting; fewer greenhouse gas emissions; no sulfur dioxide from roasting sulfide ores	Bioleaching leachate can be toxic (acidic, contains metal ions) and may pollute groundwater
Phytoextraction can clean up (remediate) contaminated land	Small yield of metal per run; the metal compound still needs further processing

Exam Tip: "Evaluate" questions need **both sides** and a **judgement**. A good finishing line: "Biological methods are better for low-grade ores where traditional extraction is uneconomic, even though they are slow."

4.7 Recycling Metals

Metal ores are **finite** resources. Recycling metals means collecting used items, sorting them, and melting them down to make new products.

Advantages of recycling	Disadvantages / issues
Conserves finite metal ores, so supplies last longer	Collection and transport of scrap uses energy and costs money
Usually uses much **less energy** than extracting from ore (e.g. recycling aluminium uses about 5% of the energy of electrolysis)	Sorting mixed metals and removing coatings is labour-intensive
Less mining → less damage to habitats, less noise, dust and waste rock	Some items are hard to separate (e.g. alloys, metals bonded to plastics)
Less waste going to **landfill**	The quality can be lower if metals are contaminated
Economic benefits: saves money (cheaper energy, fewer imports), creates jobs in collection and processing	May not be economic if the metal is cheap and abundant or the price of scrap is low

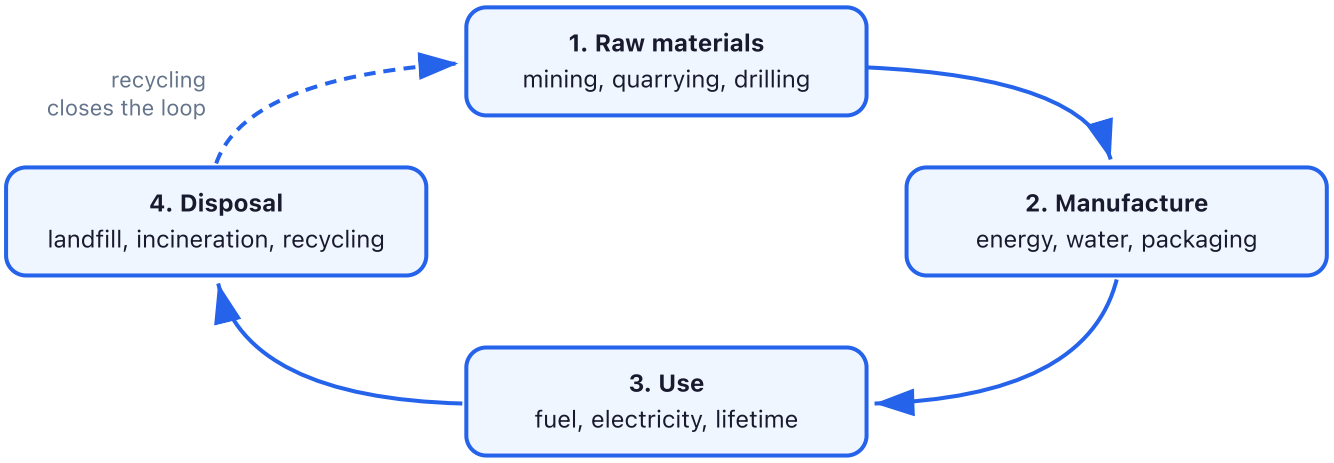
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Exam Tip: Recycling aluminium is a favourite example because aluminium extraction by electrolysis uses so much electricity. Link the saving in energy to **both** lower cost **and** lower CO_2 emissions (if electricity is generated from fossil fuels).

4.8 Life Cycle Assessments

A **life cycle assessment (LCA)** considers the effect on the environment of a product at each stage of its life: (1) **obtaining the raw materials**, (2) **manufacturing** the product, (3) **using** the product, and (4) **disposing** of the product when it is no longer useful.

The four stages of a life cycle assessment



At each stage an LCA looks at the use of **energy**, **water** and other resources, and the **pollution and waste** produced (CO_2, other gases, solid waste, effluent). Some factors, such as the amount of energy used or the mass of CO_2 released, are easy to **quantify**; others, such as the visual impact of a quarry or the harm from litter, need **value judgements**, so LCAs can be subjective or even biased (e.g. selective LCAs written to support advertising claims).

Evaluating LCA data

Worked Example: The table compares a single-use plastic (polythene) carrier bag and a paper bag.

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Stage / factor	Plastic bag	Paper bag
Raw materials	Crude oil (finite)	Wood from trees (renewable)
Energy to manufacture (kJ per bag)	1500	2500
CO_2 released in manufacture (g per bag)	40	110
Times re-used	Many (strong, waterproof)	Usually once
Disposal	Not biodegradable; long-lasting in landfill/litter; can be recycled	Biodegradable; easily recycled

Evaluation: the paper bag uses *more* energy (2500 vs 1500 kJ, 1.7 × as much) and releases nearly three times as much CO_2 in manufacture, but comes from a renewable resource and biodegrades. The plastic bag uses a finite resource and persists in the environment, but has lower manufacturing impacts and can be re-used many times. **Judgement:** if the plastic bag is re-used several times and then recycled, it probably has the lower overall impact; the conclusion depends on how the bags are actually used and disposed of.

Exam Tip: In LCA questions always (1) **quote numbers** from the data and compare them (e.g. "twice as much"), (2) consider **every stage**, and (3) end with a **conclusion** that is supported by the data. Mention that not all effects can be measured objectively.

4.9 Reversible Reactions

A **reversible reaction** is one in which the products can react to re-form the original reactants. It is shown by the symbol ⇌. The direction of some reversible reactions can be

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changed by altering the **conditions** (e.g. temperature).

$$A + B \rightleftharpoons C + D$$

Examples

- **Ammonium chloride:** heating the white solid decomposes it into ammonia and hydrogen chloride gases; at the cooler top of the tube these recombine to form white solid again.

  $$NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$$

- **Hydrated copper(II) sulfate:** heating blue crystals drives off the water of crystallisation, leaving white anhydrous copper(II) sulfate; adding water turns it blue again (and releases heat). This is also used as a **test for water**.

  $$CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(l)$$

  blue white

- **The Haber process:** $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

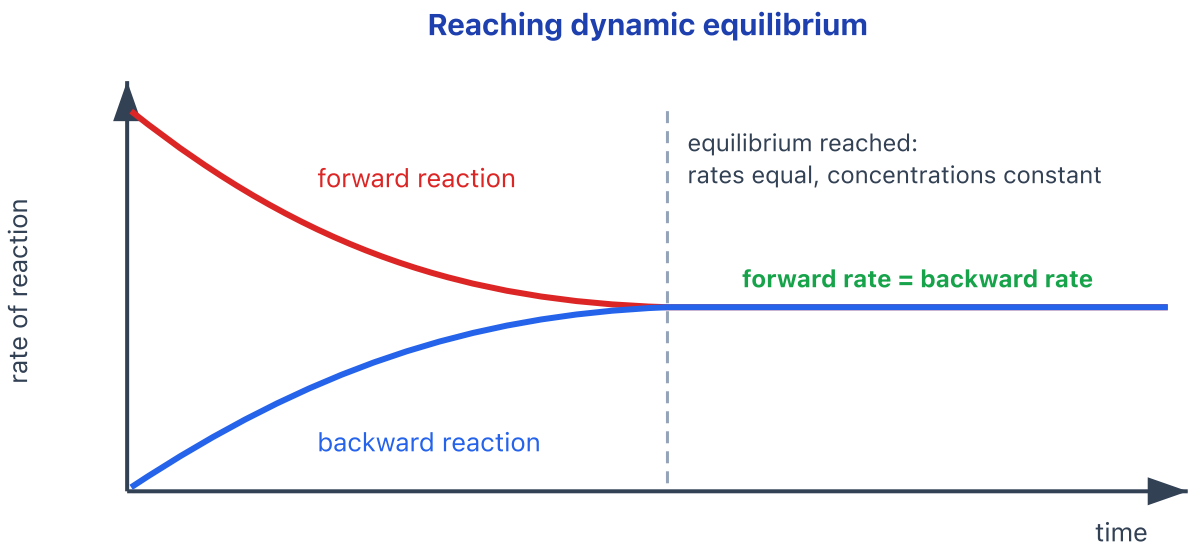
If a reversible reaction is **exothermic** in one direction, it is **endothermic** in the other, and the same amount of energy is transferred each way.

4.10 Dynamic Equilibrium

A reversible reaction in a **closed system** (nothing can get in or out) reaches **dynamic equilibrium** when the **forward and backward reactions are happening at the same rate**. The reactions have not stopped, but the **concentrations of reactants and products remain constant**.

At the start only reactants are present, so the forward reaction is fast and the backward reaction is zero. As reactants are used up, the forward rate falls; as products build up, the backward rate

rises. Eventually the two rates become equal.



Exam Tip: Two common errors: (1) saying "the reaction has stopped" at equilibrium — it hasn't, both reactions continue (that is what *dynamic* means); (2) saying "the concentrations of reactants and products are equal" — they are **constant**, not necessarily equal.

4.11 The Haber Process

Ammonia is made by the **Haber process**, a reversible reaction between nitrogen and hydrogen that reaches a dynamic equilibrium:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \text{(forward reaction exothermic)}$$

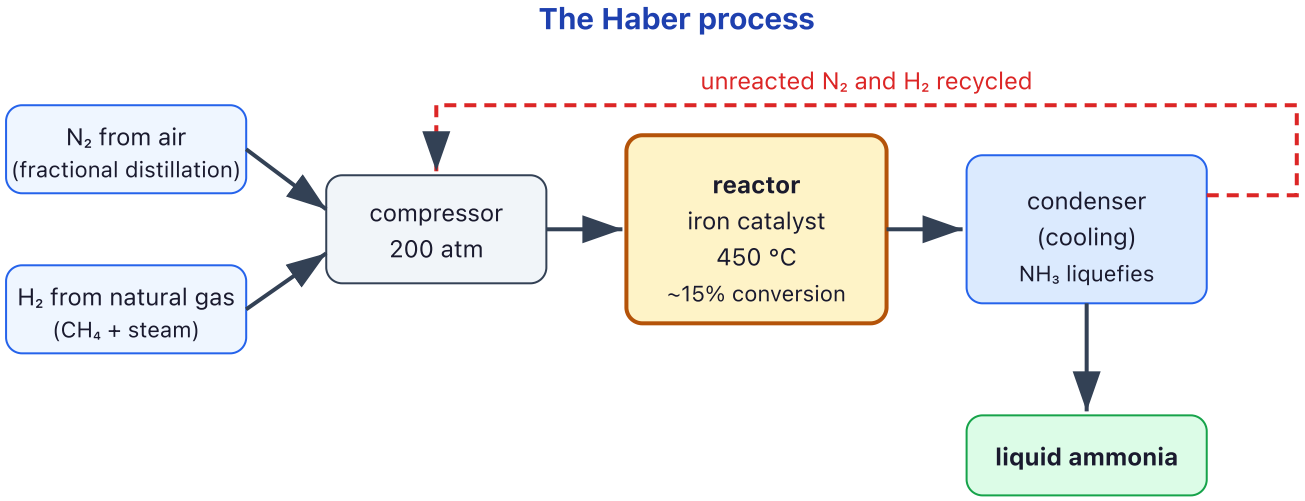
Raw materials

- **Nitrogen** is extracted from the **air** (about 78% nitrogen) by fractional distillation of liquefied air.
- **Hydrogen** is obtained from **natural gas** (methane), by reacting it with steam.

Conditions you must recall

Condition	Value

Temperature	**450 °C**
Pressure	**200 atmospheres**
Catalyst	**Iron**



The gases leaving the reactor are cooled. Ammonia has a much higher boiling point than nitrogen and hydrogen, so it **liquefies** and is removed; the unreacted nitrogen and hydrogen are **recycled** back into the reactor, so very little is wasted even though only a fraction reacts on each pass.

> **Exam Tip:** Ammonia is used mainly to make **fertilisers** (e.g. ammonium nitrate). The detail of why 450 °C and 200 atm are compromise conditions, and how fertilisers are made, is chemistry-only content in Topic 5 — but the equilibrium reasoning below is needed for both.

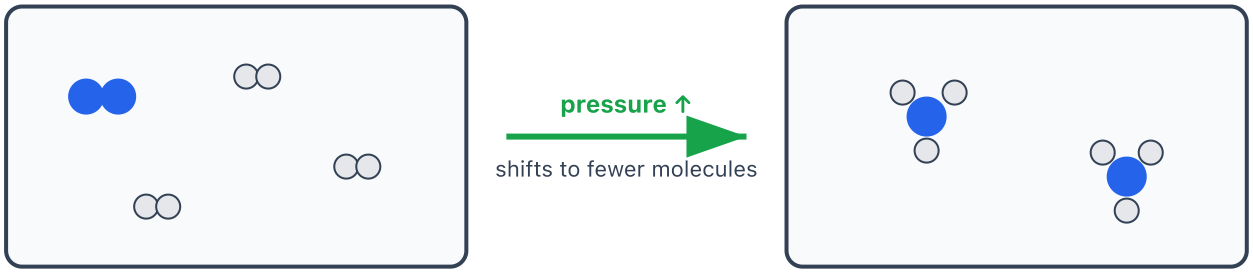
4.12 Changing the Position of Equilibrium (HT)

When a system at equilibrium is changed, the **position of equilibrium shifts** to **oppose the change**. ("Position of equilibrium" just means the relative amounts of reactants and products.) This lets you **predict** the effect of changing temperature, pressure and concentration.

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Change made	The equilibrium shifts...	Effect on $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (forward exothermic)
Increase **temperature**	in the **endothermic** direction (to absorb the extra heat)	Shifts **left** — less NH_3 (lower yield)
Decrease temperature	in the **exothermic** direction	Shifts **right** — more NH_3
Increase **pressure** (gases)	towards the side with **fewer molecules of gas**	4 gas molecules on left, 2 on right → shifts **right** — more NH_3
Decrease pressure	towards the side with **more** gas molecules	Shifts **left**
Increase **concentration** of a reactant	to use up the added substance (to the right)	Adding N_2 or H_2 → shifts **right** — more NH_3
Remove a product	to replace it (to the right)	Removing NH_3 → shifts **right**
Add a **catalyst**	**No shift** — speeds up forward and backward reactions equally	Equilibrium reached sooner; same yield

Why higher pressure favours ammonia



$N_2 + 3H_2$: **4 gas molecules**

$2NH_3$: **2 gas molecules**

Worked Example: (HT) Hydrogen and iodine react in a sealed container: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$. The forward reaction is exothermic. Predict the effect on the amount of HI of (a) raising the temperature, (b) raising the pressure.

(a) Raising the temperature favours the **endothermic** direction, which is the **backward**

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reaction. The equilibrium shifts left, so the amount of **HI decreases**.
(b) There are 2 molecules of gas on the left (1 H_2 + 1 I_2) and 2 on the right. Because the number of gas molecules is the **same on both sides**, changing pressure has **no effect** on the position of equilibrium (although the rate increases).

Worked Example: (HT) Nitrogen dioxide (brown) and dinitrogen tetroxide (colourless) are in equilibrium: $2NO_2(g) \rightleftharpoons N_2O_4(g)$. When a sealed syringe of the mixture is placed in ice, the colour becomes paler. Deduce whether the forward reaction is exothermic or endothermic.

Paler colour → less NO_2 → equilibrium has shifted **right** on cooling. Lowering the temperature favours the **exothermic** direction, so the **forward reaction is exothermic**.

Exam Tip: Structure every equilibrium answer in three steps: **(1)** state the change; **(2)** state which direction is favoured and *why* (endothermic/exothermic, fewer/more gas molecules, use up added substance); **(3)** state the effect on the yield of the named product. Always count gas molecules using the **balancing numbers** and ignore solids and liquids.

Key Points Summary

- **4.1, 4.3** — Recall the series K, Na, Ca, Mg, Al, (C), Zn, Fe, (H), Cu, Ag, Au; reactivity = tendency to form positive ions. Deduce an order from reactions with water, dilute acid and displacement.
- **4.2 (HT)** — Displacement reactions are redox: the more reactive metal loses electrons (oxidised), the less reactive metal ion gains electrons (reduced). Write ionic and half equations, leaving out spectator ions.
- **4.5, 4.6** — Oxidation = gain of oxygen; reduction = loss of oxygen. Carbon and carbon monoxide are reducing agents.
- **4.4** — Most metals are extracted from ores; unreactive metals such as gold are found uncombined.
- **4.7** — Below carbon → heat with carbon (iron in the blast furnace); above carbon → electrolysis of the molten compound (aluminium from Al_2O_3 in cryolite, carbon anodes replaced).

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- **4.8 (HT)** — Evaluate bioleaching and phytoextraction for low-grade ores: less environmental damage and energy, but slow.
- **4.9** — Metals lower in the series are more resistant to oxidation (gold does not tarnish).
- **4.10** — Recycling conserves finite ores and energy, reduces landfill and mining damage; weigh up the costs of collection and sorting.
- **4.11, 4.12** — An LCA covers raw materials, manufacture, use and disposal; evaluate LCA data using numbers and reach a conclusion.
- **4.13** — Reversible reactions use $\rightleftharpoons$; conditions can change their direction (NH_4Cl, hydrated $CuSO_4$).
- **4.14** — Dynamic equilibrium (closed system): forward rate = backward rate; concentrations constant.
- **4.15, 4.16** — Haber process: N_2 (from air) + $3H_2$ (from natural gas) $\rightleftharpoons$ $2NH_3$; 450 °C, 200 atm, iron catalyst.
- **4.17 (HT)** — Predict shifts in equilibrium position with temperature (favours endothermic when raised), pressure (favours fewer gas molecules when raised) and concentration; a catalyst does not change the position.

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